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orbital is spherical in nature. The shape of p orbital is spherical and the d orbital has clover leaf shape.

Electrons are filled in various orbitals in the increasing order of their energies which is known as Aufbau principle. The relative energies of various orbitals are given by (n+l) rule which states that, the lower the value of (n + l) for an orbital, the lower is its energy. If two orbitals have the same value of (n + l), the orbital with lower value of n will have the lower energy. As per Pauli's exclusion principle, the maximum number of electrons that can be accommodated in an orbital is two. In the case of degenerate orbitals electron pairing does not take place until all the available degenerate orbitals contain one electron each. This is known as Hund's rule. Based on these principles, electronic configurations of atoms can be written. In degenerate orbitals, the completely filled and half filled configurations are more stable than the partially filled configurations. This is due to the symmetry and exchange energies.